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DEVICE SETUP GUIDE

Garden Monitor

Sensors, Watering & Factory Reset Reference

This guide covers how to add sensors and watering relay systems to your Garden Monitor, and how to factory reset each device if needed.

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System Overview

Your Garden Monitor system consists of:

Device	Purpose
Garden Monitor	The central hub. Connects to your WiFi and provides the web interface for controlling everything.
Indoor Sensor	Monitors soil moisture, temperature, and humidity for indoor plants.
Outdoor Sensor	Monitors soil moisture for outdoor garden beds.
Relay Watering System	Controls up to 4 watering zones via solenoid valves or pumps.

The Garden Monitor communicates wirelessly with sensors and watering systems using a long-range radio (LoRa). All sensors and watering systems must be **paired** to your Garden Monitor before they can be used.

Accessing the Garden Monitor

1. Make sure your Garden Monitor is powered on and connected to your WiFi network.
2. On a device connected to the same WiFi network, open a web browser.
3. Go to: **http://gardenmonitor.local**
 - If this doesn't work, check your router's device list for the IP address of "gardenmonitor" and use that instead (e.g. <http://192.168.1.50>).
4. The Garden Monitor dashboard will load. From here, navigate to **Settings** to manage devices.

Adding an Indoor or Outdoor Sensor

Step 1 — Power on the sensor

Insert the batteries (or connect power) to the sensor. The sensor will automatically begin broadcasting to the Garden Monitor, and will appear in the **Pending Devices** tab within a minute or two.

Tip: Keep the sensor within a reasonable distance of the Garden Monitor during setup to ensure a reliable signal.

Step 2 — Pair the sensor

1. In the web interface, go to **Settings**.
2. Click the **Pending Devices** tab.
3. Your sensor will appear in the **Pending Sensors** list, showing its device ID and current readings.
4. Click **Pair & Activate**.
5. The Garden Monitor will send a pairing request to the sensor. Wait up to 30 seconds for the pairing to complete.
6. Once paired, the sensor will automatically enter **calibration mode** — the sensor display will update to show a calibration link.

If the sensor does not appear after a few minutes, check the battery and make sure you are within radio range of the Garden Monitor.

Step 3 — Calibrate the sensor

Calibration teaches the sensor what "dry" and "wet" look like for your specific soil. You will need to do this once per sensor.

1. After pairing, click the **Calibrate** link shown next to the sensor.
2. **Dry calibration:**
 - Hold the sensor probe in the **air** (not touching anything).
 - Wait for the live reading to appear and stabilise.
 - Click **Set Dry Value**.
3. **Wet calibration:**
 - Insert the sensor probe into a **glass of water**.
 - Wait for the live reading to appear and stabilise.
 - Click **Set Wet Value**.
4. Calibration is now complete. The sensor will be added to your active sensors list and the page will redirect automatically.

After wet calibration, dry the sensor probe before placing it in soil.

Step 4 — Configure the sensor

1. Go to **Settings** → **Sensors** tab.
2. Find your newly added sensor and set:
 - **Name** — a descriptive name (e.g. "Herb Garden", "Living Room Fiddle Leaf")
 - **Location** — Indoor or Outdoor
 - **Soil Type** — Regular Soil or Potting Mix
 - **Moisture Preference** — Dry, Normal, or Wet (based on your plant's needs)
 - **Garden** — assign it to one of your gardens if you have multiple set up
3. Your sensor will now appear on the dashboard and begin reporting readings.

Adding a Relay Watering System

The Relay Watering System controls up to **4 watering zones** (relays). Each zone can be independently named, enabled, and assigned to a garden.

Step 1 — Power on the watering system

Connect power to the Relay Watering System. It will immediately begin broadcasting to the Garden Monitor and will appear in the **Pending Devices** tab within a minute or two.

Make sure your solenoids or pumps are **not** connected to the relay outputs during initial setup. Connect them after pairing and configuration is complete.

Step 2 — Pair the watering system

1. In the web interface, go to **Settings**.
2. Click the **Pending Devices** tab.
3. Your watering system will appear in the **Pending Watering Systems** list, showing its device ID.
4. Optionally, enter a **name** for the system before pairing.
5. Click **Pair & Adopt**.
6. The Garden Monitor will pair with the watering system. This is usually completed within a few seconds.
7. Once paired, the relay system will appear as active on your dashboard.

Step 3 — Configure the relays

1. Go to **Settings** → **Watering** tab.
2. Find your newly paired watering system.
3. For each of the 4 relay channels, you can set:
 - **Relay Name** — e.g. "Front Lawn", "Veggie Bed", "Drip Line"
 - **Garden** — assign the relay to one of your gardens
 - **Enable/Disable** — toggle individual relays on or off
4. Your watering system will now be controllable from the dashboard. You can manually trigger each zone and set durations.

Factory Resetting Devices

A factory reset clears the pairing information from a device, returning it to a state where it can be re-paired. Use this if you need to move a device to a different Garden Monitor, or if a device is behaving incorrectly.

Note: A factory reset only clears pairing — it does not change any hardware settings.

Factory Reset — Indoor Sensor

The indoor sensor uses a **power-cycle counter** to detect a reset request.

1. **Power off** the sensor (remove batteries or disconnect power).
2. **Power it back on**, then immediately **power it off** again.
3. Repeat this **3 times in quick succession** (on–off–on–off–on).
4. After the third power-on, the sensor will clear its pairing and return to unpaired discovery mode.

Timing is important. Each power cycle must happen within about 30 seconds. If you wait too long between cycles, the counter resets and you'll need to start again.

Alternatively, if the sensor is already paired and within radio range, the Garden Monitor can send a remote reset. Go to **Settings** → **Sensors**, find the sensor, and click **Reset**.

Factory Reset — Outdoor Sensor

The outdoor sensor has a **BOOT button** on the device board.

1. **Locate the BOOT button** on the sensor board.
2. **Hold the BOOT button down** while powering the sensor on (connecting the battery or power).
3. **Keep holding** the button for **3 seconds**.
 - Release the button any time before 3 seconds to cancel the reset.
4. After 3 seconds, the sensor will clear its pairing and restart in unpaired discovery mode.

Alternatively, if the sensor is already paired and within radio range, the Garden Monitor can send a remote reset. Go to **Settings** → **Sensors**, find the sensor, and click **Reset**.

Factory Reset — Relay Watering System

The relay watering system has a **factory reset button** (a push button connected between pin 8 and ground on the device board).

1. **Locate the reset button** on the relay board.
2. **Hold the reset button down** while powering the system on.
3. The LED on the device will **blink 8 times** to confirm the reset was successful.
4. The watering system will restart in unpaired discovery mode, ready to be re-paired.

The device's unique ID (e.g. WS04821) is never erased — only the pairing information is cleared.

Factory Reset — Garden Monitor (Hub)

Warning: This will erase **all** configuration including paired sensors, watering systems, gardens, WiFi settings, and historical data. This cannot be undone.

Method A — Web interface (recommended):

1. Go to **Settings** → **System** tab.
2. Scroll to the **Factory Reset** section.
3. Click **Factory Reset Device**.
4. Confirm the action in the dialog box.
5. The Garden Monitor will restart with factory defaults.

Method B — Hardware button:

1. Locate the **BOOT button** on the Garden Monitor board.
2. While the device is powered on, **hold the BOOT button for 5 seconds**.
3. Release the button when prompted (serial console shows countdown).
4. The device will restart with factory defaults.

After a factory reset, you will need to reconnect the Garden Monitor to your WiFi and re-pair all sensors and watering systems from scratch.

Troubleshooting

Problem	What to try
Sensor doesn't appear in Pending Devices	Check batteries. Move sensor closer to the Garden Monitor. Wait 2–3 minutes.
Pairing times out	Ensure the sensor is within range. Try removing and reinserting batteries to restart the sensor, then try again.
Calibration link doesn't appear after pairing	Refresh the Settings page. If the sensor shows as "Calibrating", click directly on its ID to open the calibration page.
Calibration page shows no live readings	The sensor may have lost radio contact. Click Retry on the calibration page.
Watering relay doesn't appear in Pending Devices	Check power to the relay board. Move it closer to the Garden Monitor.
Relay not responding to commands	Check that the relay is shown as online on the dashboard. If offline, check power and proximity.
Factory reset not working on indoor sensor	Make sure each power cycle is within 30 seconds of the previous one. The counter clears if the device stays on for more than 30 seconds.
Factory reset not working on outdoor sensor	Hold the BOOT button for the full 3 seconds — releasing early cancels the reset.
Can't access gardenmonitor.local	Try the IP address directly. Check that your phone/computer is on the same WiFi network as the Garden Monitor.